TrES-2b

R-1612

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_140707` · created 2026-08-02T03:29:54+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456845.78288 +/- 0.0003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.043
Transit depth (Rp/Rs)^2	2.22 +/- 1.27 [%]
Orbital Inclination (inc)	79.07 +/- 2.9
Airmass coefficient 1 (a1)	20.7 +/- 12.25
Airmass coefficient 2 (a2)	-2.85 +/- 1.29
Scatter in the residuals of the lightcurve fit is	56.01 %
Best Comparison Star	#2 - [74, 133]
Optimal Aperture	2.1
Optimal Annulus	8.65
Transit Duration (day)	0.012 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.043 vs 0.1254 (deviation 0.48σ)
Duration fit vs published	0.012 d vs 0.0746 d (deviation 2.12σ)
Mid-time O–C	0.5 min
Depth SNR	3.1
Residual scatter	56.01 %

Light curve

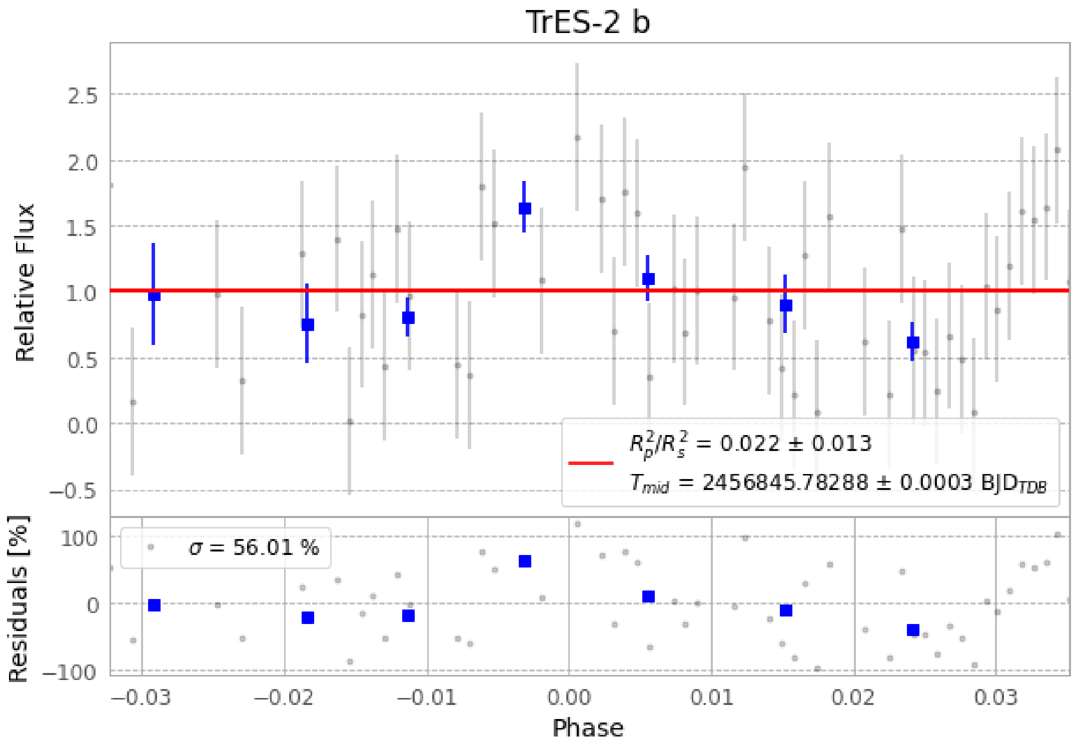

Figure 1 — FinalLightCurve_TrES-2 b_2014-07-06.png (SHA-256 ed6e196ceb2e8171…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 262], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1d3ff53dbe62bfc8…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

81 FITS frames (53 MB), `TRES-2140707044813.FITS` → `TRES-2140707084812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140707.log (dbf3027b7a5c…), FinalLightCurve_TrES-2 b_2014-07-06.png (ed6e196ceb2e…), FinalLightCurve_TrES-2 b_2014-07-06.csv (031dbbf101eb…)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)